

Excretory Products and Their Elimination

Complete Exam-Oriented Master Notes with Diagrams, Formulas, Counter-Current Mechanism, and Solved NCERT Exercises.

Weightage: **High (NEET/Boards)**

Key Focus: **Nephron & RAAS**

Osmolarity Range: **300–1200 mOsmol L⁻¹**

QUICK NAV:

[16.1 Waste Types](#)[16.1 Excretory Organs](#)[16.1 Human System](#)[16.2 Urine Formation](#)[16.3 Tubule Functions](#)[16.4 Counter-Current](#)[16.5 Regulation](#)[16.6 Micturition](#)[16.7 Other Organs](#)[16.8 Disorders](#)[Exercise Solutions](#)



CHAPTER 16 VISUAL MIND MAP

Architectural Overview

1. Nitrogenous Wastes

Modes

- **Ammonotelism:** Most toxic, high H₂O needed (Bony fish, aquatic larvae)
- **Ureotelism:** Less toxic, urea in liver (Mammals, marine fish)
- **Uricotelism:** Least toxic, pellet/paste, min H₂O (Birds, reptiles, insects)

2. Human Urinary System

Structure

- **Kidneys:** Pair, T12–L3 level, 120–170g, Cortex & Medulla
- **Nephron Unit:** Malpighian Body (Glomerulus + Bowman's) + Renal Tubule
- **Nephron Types:** Cortical (85%, short loop) vs Juxtamedullary (15%, long loop + Vasa Recta)

3. Urine Formation & Flow

Physiology

- **Glomerular Ultrafiltration:** GFR = 125 mL/min (180 L/day) via 3 layers
- **Reabsorption:** 99% reabsorbed (PCT reabsorbs 70–80% electrolytes)
- **Counter-Current Mechanism:** Loop of Henle & Vasa Recta build gradient (300→1200 mOsmol L⁻¹)

4. Regulation & Disorders

Control

- **ADH / Vasopressin:** Water reabsorption in DCT/CD
- **JGA & RAAS Axis:** Renin → Angiotensin II → Aldosterone
- **ANF Check:** Vasodilation lowers BP (opposes RAAS)
- **Pathologies:** Uremia, Hemodialysis, Calculi, Glomerulonephritis

Metabolic Accumulation & Excretion Concept

Animals accumulate **ammonia, urea, uric acid, carbon dioxide, water** and ions like **Na⁺, K⁺, Cl⁻, phosphate, sulphate**, etc., either by metabolic activities or by excess ingestion. These nitrogenous products are derived mainly from protein and nucleic acid catabolism and must be eliminated from the body totally or partially.

1. Ammonotelism

Max Toxicity

- **Excretory Product:** Ammonia (NH₃)
- **Toxicity:** Extremely toxic; requires large amounts of water for elimination.
- **Excretion Mode:** Readily soluble; excreted by diffusion across body surface or gill surfaces as **ammonium ions (NH₄⁺)**.
- **Kidney Role:** Kidneys play *no significant role* in its removal.

Textbook Examples: Many bony fishes, aquatic amphibians, and aquatic insects.

2. Ureotelism

Moderate Toxicity

- **Excretory Product:** Urea
- **Adaptation:** Terrestrial adaptation for water conservation.
- **Formation Site:** Ammonia produced by metabolism is converted into **urea in the LIVER** and released into blood.
- **Matrix Retention:** Some urea may be retained in kidney matrix to maintain desired osmolarity.

Textbook Examples: Mammals, many terrestrial amphibians, and marine fishes.

3. Uricotelism

Least Toxicity

- **Excretory Product:** Uric acid
- **Physical State:** Excreted as **pellet or paste** with minimum loss of water.
- **Toxicity:** Least toxic among all three nitrogenous wastes.

- **Water Economy:** Ideal for maximum conservation of body water.

Textbook Examples: Reptiles, birds, land snails, and insects.

MASTER COMPARISON TABLE: MODES OF NITROGENOUS EXCRETION

Feature / Parameter	Ammonotelism	Ureotelism	Uricotelism
Major Waste Product	Ammonia (NH ₃) / Ammonium ions (NH ₄ ⁺)	Urea	Uric acid
Toxicity Level	Highest toxicity	Moderate toxicity (~100,000x less toxic than NH ₃)	Lowest toxicity
Water Required for Removal	Very Large volume (500 mL H ₂ O / g N)	Moderate volume (50 mL H ₂ O / g N)	Minimum loss of water (5 mL H ₂ O / g N)
Synthesis Site & Process	Direct product of deamination; no synthesis needed	Synthesized in Liver via Urea / Ornithine Cycle	Synthesized from purines in liver & kidneys
Excretory Mechanism	Diffusion across body/gill surfaces; kidneys insignificant	Filtered & excreted by Kidneys	Excreted as insoluble pellet or paste
Animal Examples	Bony fishes, aquatic amphibians, aquatic insects	Mammals, terrestrial amphibians, marine fishes	Reptiles, birds, land snails, insects



EASY WAY TO REMEMBER ANIMAL WASTE TYPES:

Aqua-living = **A**mmonia (Ammonotelic) | **U** and me (Human Mammals) = **U**rea (Ureotelic) | **P**ellet / **P**aste = **U**ric acid (Uricotelic - Birds, Reptiles, Bugs)

2 DIVERSITY OF EXCRETORY STRUCTURES ACROSS THE ANIMAL KINGDOM

NCERT Page 206

Simple Tubular Structures (Invertebrates)

1. Protonephridia / Flame Cells:

Excretory structures in **Platyhelminthes** (Flatworms e.g., *Planaria*), **rotifers**, **some annelids**, and the cephalochordate **Amphioxus**. Primarily concerned with ionic and fluid volume regulation (osmoregulation).

2. Nephridia:

Tubular excretory structures of **earthworms** and other annelids. Help to remove nitrogenous wastes and maintain fluid/ionic balance.

Complex Tubular Organs (Insects, Crustaceans & Vertebrates)

3. Malpighian Tubules:

Excretory structures of most insects including **cockroaches**. Help in removal of nitrogenous wastes and osmoregulation.

4. Antennal Glands / Green Glands:

Perform excretory function in crustaceans like **prawns**.

5. Kidneys:

Complex tubular organs present in **Vertebrates** for filtration, reabsorption, secretion, and osmoregulation.

Excretory Structure	Animal Groups / Examples	Primary Functions
Protonephridia (Flame Cells)	Platyhelminthes (<i>Planaria</i>), Rotifers, Some Annelids, <i>Amphioxus</i> (Cephalochordata)	Osmoregulation (ionic & fluid volume balance)
Nephridia	Earthworms and other Annelids	Excretion of N-wastes & fluid-ionic balance
Malpighian Tubules	Most Insects (including Cockroach)	Nitrogenous waste removal & osmoregulation
Antennal Glands / Green Glands	Crustaceans (e.g., Prawns)	Excretion & osmoregulation
Kidneys	All Vertebrates (Fishes, Amphibians, Reptiles, Birds, Mammals)	Excretion of N-wastes, osmoregulation, acid-base homeostasis

3 16.1 HUMAN EXCRETORY SYSTEM — ANATOMY & STRUCTURE

NCERT Page 206-208

1 Pair of Kidneys

Reddish brown, bean-shaped, located between last thoracic (T12) and 3rd lumbar (L3) vertebrae close to dorsal abdominal wall.

1 Pair of Ureters

Muscular tubes originating from renal pelvis through hilum; convey urine to urinary bladder.

1 Urinary Bladder

Muscular sac located in pelvic cavity; stores urine until voluntary micturition signal.

1 Urethra

Duct conducting urine from bladder to external urethral orifice for discharge.

Kidney Dimensions & External Features

Length
10 – 12 cm

Width
5 – 7 cm

Thickness
2 – 3 cm

Average Weight
120 – 170 g

- **Hilum:** A notch towards the center of the inner concave surface through which ureter, blood vessels, and nerves enter.
- **Renal Pelvis:** Broad funnel-shaped space inner to hilum with projections called **calyces** (singular: calyx).
- **Renal Capsule:** Outer tough fibrous protective layer.

Internal Zonation of Kidney

1. Outer Cortex: Outer granular layer extending between medullary pyramids as renal columns.

2. Inner Medulla: Divided into a few conical masses called **medullary pyramids** projecting into calyces.

★ **Columns of Bertini (Renal Columns):**

The cortical tissue that extends in between the medullary pyramids as renal columns.

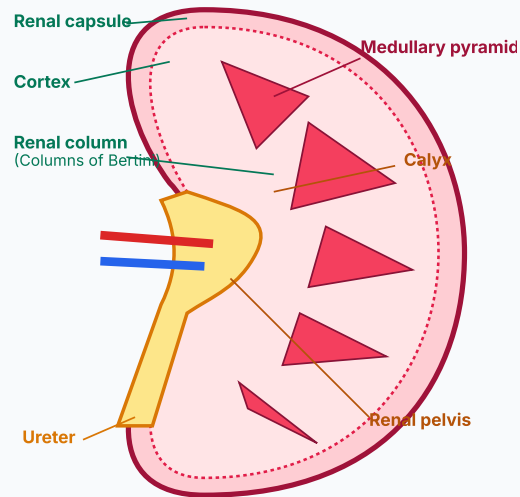


Figure 16.2 — Diagrammatic L.S. of Human Kidney

Detailed Key to Kidney Structures

- **Renal Capsule:** Tough fibrous outer membrane maintaining structural integrity.
- **Cortex:** Outer region containing Glomeruli, Bowman's capsules, PCT, and DCT.
- **Medullary Pyramids:** Conical subdivisions (usually 8–18 per kidney) containing Henle's loops and collecting ducts.
- **Calyces:** Cuplike chambers receiving urine from papillae of pyramids. Minor calyces merge into major calyces, which lead into the renal pelvis.
- **Renal Pelvis:** Funnel-like dilated proximal end of ureter.
- **Columns of Bertini:** Portions of cortex projecting into spaces between medullary pyramids.

4 STRUCTURE OF THE NEPHRON (FUNCTIONAL UNIT OF KIDNEY)

NCERT Page 207-208

Each human kidney contains nearly **1 million (10 lakh)** complex tubular structures called **nephrons**, which are the functional units of the kidney. Each nephron consists of

two main parts: **(1) Glomerulus** and **(2) Renal Tubule**.

1. Glomerulus & Malpighian Body

Filtration Unit

- **Glomerulus:** A tuft of capillary network formed by the **afferent arteriole** (a fine branch of renal artery). Blood from glomerulus is carried away by an **efferent arteriole**.
- **Bowman's Capsule:** Double-walled cup-like structure enclosing glomerulus.
- **Malpighian Body (Renal Corpuscle):** Glomerulus + Bowman's Capsule.
- **Podocytes:** Specialized epithelial cells of Bowman's capsule inner layer arranged intricately leaving minute spaces called **filtration slits or slit pores**.

2. Renal Tubule Segments

Tubular System

- **Proximal Convoluted Tubule (PCT):** Highly coiled network lined by simple cuboidal brush border epithelium (increases surface area).
- **Henle's Loop:** Hairpin-shaped tubule with **descending limb** (thin, permeable to water) and **ascending limb** (impermeable to water, transports NaCl).
- **Distal Convoluted Tubule (DCT):** Highly coiled segment located in cortex opening into collecting duct.
- **Collecting Duct:** Long straight duct running from cortex to inner medulla; DCTs of many nephrons converge into it. Opens into renal pelvis through calyces.

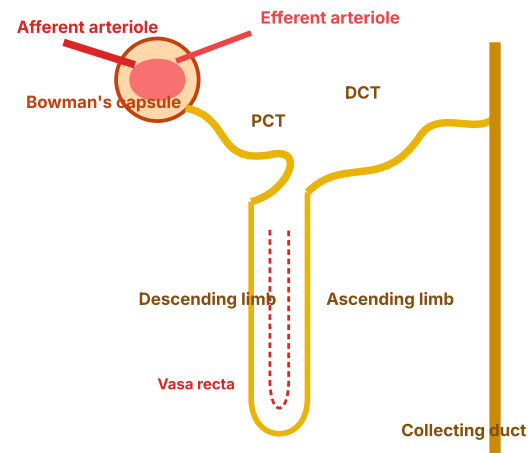


Figure 16.3 — Diagrammatic Nephron & Vasa Recta

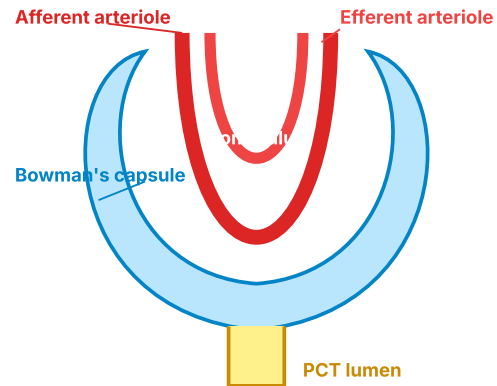


Figure 16.4 — Malpighian Body (Renal Corpuscle)

COMPARISON: CORTICAL NEPHRONS VS. JUXTAMEDULLARY NEPHRONS

Feature	Cortical Nephrons (Majority ~85%)	Juxtamedullary Nephrons (Minority ~15%)
Location of Glomerulus	Outer cortex of kidney	Inner cortex near medullary border (juxta-medullary)
Length of Loop of Henle	Very short; extends only very little into medulla	Very long; runs deep into the inner medulla
Vasa Recta Development	Vasa recta is absent or highly reduced	Vasa recta is well-developed , runs parallel to Henle's loop
Primary Function	General blood filtration and waste excretion under normal water availability	Concentration of urine via Counter-Current Mechanism during water conservation

Capillary Networks of the Nephron:

- **Peritubular Capillaries:** The efferent arteriole emerging from glomerulus forms a fine capillary network around the renal tubule.
- **Vasa Recta:** A minute vessel of peritubular network running parallel to Henle's loop forming a 'U' shaped vessel. Absent/reduced in cortical nephrons; prominent in juxtamedullary nephrons.

516.2 URINE FORMATION — THREE MAIN STEPS

NCERT Page 208-209

1. Glomerular Ultrafiltration

Step 1

- **Blood Volume Filtered:** 1100–1200 mL of blood per minute (~1/5th of cardiac output pumped by each ventricle per min).
- **Filtration Barrier (3 Layers):**
 1. Endothelium of glomerular blood vessels
 2. Epithelium of Bowman's capsule (with **Podocytes & Slit Pores**)
 3. Basement membrane between the two layers
- **Ultrafiltration Nature:** Blood is filtered so finely that almost all constituents of plasma except proteins pass into lumen of Bowman's capsule.

2. Tubular Reabsorption

Step 2

- **Massive Reabsorption:** Out of 180 Litres filtrate/day, only 1.5 L urine is released. **~99% of filtrate is reabsorbed** by renal tubules.
- **Active Transport:** Glucose, amino acids, Na⁺, etc. are reabsorbed actively against gradient.

- **Passive Transport:** Nitrogenous wastes are absorbed passively. Water reabsorption also occurs passively in initial segments.

3. Tubular Secretion

Step 3

- **Secreted Substances:** Tubular cells secrete H^+ , K^+ , and Ammonia (NH_3) into the filtrate.
- **Crucial Function:** Helps maintain **ionic balance and acid-base (pH) balance** of body fluids.

Glomerular Filtration Rate (GFR) Quantitative Values

NCERT Standard

GFR Per Minute

125 mL / minute

GFR Per Day

180 Litres / day

Actual Urine Output

1.5 Litres / day (~0.8%)

Autoregulation by JGA: Juxtaglomerular Apparatus (JGA) is a specialized region formed by cellular modifications in DCT and afferent arteriole at contact location. Fall in GFR → activates JG cells → releases **Renin** → stimulates glomerular blood flow → brings GFR back to normal.

6 16.3 FUNCTIONS OF DIFFERENT NEPHRON TUBULES

NCERT Page 209-210

1. Proximal Convoluted Tubule (PCT)

70–80% Reabsorption

- **Epithelium:** Lined by **simple cuboidal brush border epithelium** which increases surface area for reabsorption.
- **Reabsorption:** Reabsorbs **nearly all essential nutrients**, and **70–80% of electrolytes and water**.
- **Secretion & pH Homeostasis:** Maintains pH and ionic balance of body fluids by selective secretion of H^+ , K^+ , and NH_3 into filtrate, and reabsorption of HCO_3^- .

2. Loop of Henle

Medullary Gradient

- **General:** Reabsorption is *minimum* in ascending limb. Plays crucial role in maintaining high osmolarity of medullary interstitial fluid.
- **Descending Limb:** Permeable to water, almost **impermeable to electrolytes**. Concentrates filtrate as it moves down.
- **Ascending Limb: Impermeable to water**, allows transport of electrolytes (NaCl) actively or passively. Concentrated filtrate gets *diluted* moving upward.

3. Distal Convoluted Tubule (DCT)

Conditional Reabsorption

- **Reabsorption: Conditional reabsorption of Na⁺ and water** takes place here (regulated by Aldosterone & ADH). Reabsorbs HCO₃⁻.
- **Secretion:** Selective secretion of H⁺, K⁺, and NH₃ to maintain blood pH and sodium-potassium balance.

4. Collecting Duct (CD)

Final Water Concentration

- **Water Absorption:** Extends from cortex to inner medulla. Large amounts of water reabsorbed to produce **concentrated urine**.
- **Urea Diffusion:** Allows passage of small amounts of urea into medullary interstitium to maintain high osmolarity.
- **pH Balance:** Secretes H⁺ and K⁺ ions for ionic balance.

MASTER TABLE: TUBULAR TRANSPORT PROPERTIES ACROSS NEPHRON SEGMENTS

Nephron Segment	Epithelium Type	Substances Reabsorbed	Substances Secreted	Permeability Properties
PCT	Simple cuboidal brush border	All essential nutrients, 70–80% electrolytes (NaCl), H ₂ O, HCO ₃ ⁻	H ⁺ , K ⁺ , NH ₃	Highly permeable to water & solutes
Descending Loop of Henle	Simple squamous	Water (H ₂ O)	None significantly	Permeable to H₂O ; Impermeable to electrolytes
Ascending Loop of Henle	Squamous (thin) / Cuboidal (thick)	NaCl (active & passive)	Urea (enters thin segment)	Impermeable to H₂O ; Permeable to electrolytes
DCT	Simple cuboidal	Na ⁺ , H ₂ O (conditional via Aldosterone/ADH), HCO ₃ ⁻	H ⁺ , K ⁺ , NH ₃	H ₂ O permeability depends on ADH level
Collecting Duct	Simple columnar/cuboidal	Water (large amount), Urea (small amount into interstitium)	H ⁺ , K ⁺	Water permeability controlled by ADH

Core Principle of Counter-Current System

Mammals have the ability to produce urine **4 times more concentrated** than the initial filtrate (up to **1200 mOsmol L⁻¹** compared to 300 mOsmol L⁻¹ in cortex). This hyperosmolar gradient is established and maintained by two counter-current systems operating in close proximity: **(1) Loop of Henle** and **(2) Vasa Recta**.

1. Counter-Current in Loop of Henle

Filtrate flows in **opposite directions** in the two limbs of Henle's loop:

- Downward in descending limb (water moves out → filtrate concentrated)
- Upward in ascending limb (NaCl pumped out → filtrate diluted)

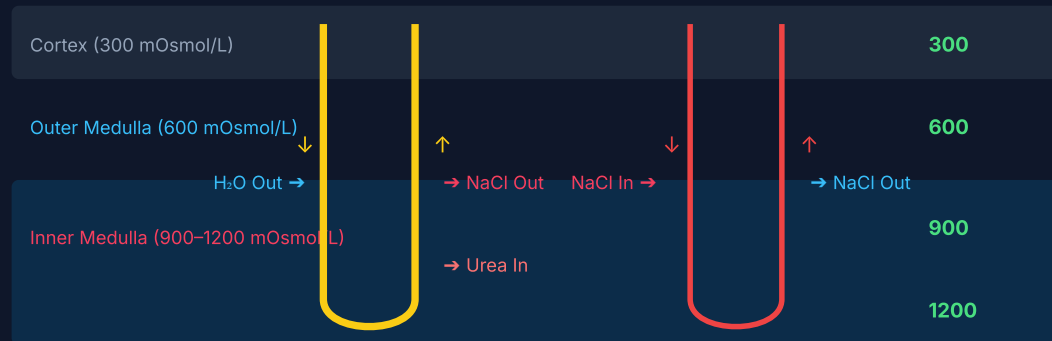
2. Counter-Current in Vasa Recta

Blood flows in **opposite directions** in the two limbs of Vasa Recta:

- Downward in descending limb (takes up NaCl from interstitium)
- Upward in ascending limb (returns NaCl back to interstitium)

Osmolarity Gradient in Medullary Interstitium

300 → 1200 mOsmol L⁻¹



Role of NaCl:

NaCl transported by ascending limb of Henle's loop is exchanged with descending limb of Vasa Recta. NaCl is returned to interstitium by ascending portion of Vasa Recta.

Role of Urea:

Small amounts of urea enter thin segment of ascending limb of Henle's loop, which is transported back to medullary interstitium by collecting tubule to complete the loop.

8 16.5 REGULATION OF KIDNEY FUNCTION (HORMONAL CONTROL)

NCERT Page 212

Kidney function is efficiently monitored and regulated by hormonal feedback mechanisms involving **Hypothalamus**, **JGA**, and to a certain extent, the **Heart**.

1. Hypothalamus & ADH Axis

Osmoreceptor Regulation

1. Fluid loss / dehydration activates **Osmoreceptors** in body.
2. Stimulates Hypothalamus to release **Antidiuretic Hormone (ADH) / Vasopressin** from neurohypophysis.
3. ADH facilitates water reabsorption from DCT and Collecting Duct → prevents **Diuresis**.
4. Constricts blood vessels → increases BP → increases Glomerular Blood Flow & GFR.
5. Increase in fluid volume switches off osmoreceptors (Negative Feedback).

★ **Key Outcome: Conserves body water & elevates GFR.**

2. Renin-Angiotensin Mechanism (RAAS)

JGA Endocrine Axis

1. Fall in Glomerular Blood Flow / BP / GFR activates JG cells.
2. JG cells release enzyme **Renin** into blood.
3. Renin converts **Angiotensinogen** → **Angiotensin I** → **Angiotensin II**.
4. **Angiotensin II** (powerful vasoconstrictor) increases BP and GFR.
5. Angiotensin II activates **Adrenal Cortex** to release **Aldosterone**.
6. Aldosterone causes reabsorption of **Na⁺** and **H₂O** from distal tubules.

★ **Key Outcome: Restores Blood Pressure & GFR back to normal.**

3. Atrial Natriuretic Factor (ANF)

Heart Feedback Check

1. Increase in blood flow to atria of heart stimulates atrial wall.
2. Atria release **Atrial Natriuretic Factor (ANF)**.
3. ANF causes **Vasodilation** (dilation of blood vessels).
4. Decreases blood pressure.
5. **ANF mechanism acts as a check / antagonist to the Renin-Angiotensin mechanism.**

★ **Key Outcome: Lowers BP; counteracts RAAS overactivation.**

RAAS CASCADE FLOW FORMULA:

Fall in GFR/BP → JG Cells release Renin → Angiotensinogen → Angiotensin I → Angiotensin II → Aldosterone (Adrenal) →
Na⁺ & H₂O Reabsorption

The Micturition Reflex Pathway

Urine formed by nephrons is stored in urinary bladder until voluntary CNS signal:

1. Filling of urinary bladder with urine causes **stretching** of bladder walls.
2. **Stretch receptors** on bladder walls send sensory signals to Central Nervous System (CNS).
3. CNS passes motor messages to initiate simultaneous:
 - **Contraction of smooth muscles of bladder**
 - **Relaxation of urethral sphincter**
4. Leads to release of urine (**Micturition** via Micturition Reflex).

Normal Physical & Chemical Parameters

Daily Volume

1.0 – 1.5 Litres / day

pH Value

Slightly Acidic (pH ~ 6.0)

Color & Fluid

Light yellow watery fluid

Urea Excreted / Day

25 – 30 grams / day

Clinical Diagnostic Indicators in Urine:

- **Glycosuria:** Presence of glucose in urine.
 - **Ketonuria:** Presence of ketone bodies in urine.
- Both Glycosuria and Ketonuria are indicative of Diabetes Mellitus.

Other than kidneys, **lungs, liver, and skin** also assist significantly in the elimination of metabolic excretory wastes.

1. Lungs

CO₂ Elimination

- Remove large amounts of **CO₂** (approximately 200 mL / minute or ~18 L/hour).
- Eliminate significant quantities of **water in the form of vapor** every day.

2. Liver (Largest Gland)

Bile Secretions

- Secretes bile-containing substances: **bilirubin, biliverdin, cholesterol, degraded steroid hormones, vitamins, and drugs**.
- Most pass out along with digestive wastes in feces.

3. Skin & Salivary Glands

Sweat & Sebum

- **Sweat Glands:** Produce sweat containing **NaCl, small amounts of urea, lactic acid**. Primary role: cooling effect.
- **Sebaceous Glands:** Eliminate **sterols, hydrocarbons, and waxes** through sebum (protective oily covering).
- **Saliva:** Small amounts of nitrogenous wastes can be eliminated through saliva.

Organ	Excretory Secretion / Product	Key Excretory Components Removed
Lungs	Expired Air	CO ₂ (~200 mL/min) and water vapor
Liver	Bile	Bilirubin, Biliverdin, Cholesterol, Degraded steroid hormones, Vitamins, Drugs
Sweat Glands (Skin)	Sweat	NaCl, small amounts of Urea, Lactic acid
Sebaceous Glands (Skin)	Sebum	Sterols, Hydrocarbons, Waxes
Salivary Glands	Saliva	Trace amounts of nitrogenous wastes

11 16.8 DISORDERS OF THE EXCRETORY SYSTEM

NCERT Page 213-214

1. Uremia & Hemodialysis (Artificial Kidney)

Critical Condition

Uremia: Malfunctioning of kidneys leading to accumulation of high levels of urea in blood; highly harmful and may lead to kidney failure.

Step-by-Step Procedure of Hemodialysis:

- 1. Blood drained from a **convenient artery** is pumped into a dialysing unit (artificial kidney).
- 2. An anticoagulant like **HEPARIN** is added to the drained blood.
- 3. The dialysing unit contains a **coiled cellophane tube** surrounded by dialysing fluid having *same composition as plasma except NO nitrogenous wastes*.
- 4. Porous cellophane membrane allows nitrogenous wastes to freely diffuse down concentration gradient into dialysing fluid, clearing the blood.
- 5. Cleared blood is pumped back into patient's body through a **VEIN** after adding **ANTI-HEPARIN**.

2. Kidney Transplantation

- Ultimate method in correction of acute renal failure (kidney failure).
- A functioning kidney from a donor, **preferably a close relative**, is transplanted to minimize chances of immune rejection by host system.

3. Calculi & Inflammatory Disorders

Renal Calculi: Stone or insoluble mass of crystallised salts (**oxalates**, etc.) formed within the kidney.

Glomerulonephritis: Inflammation of glomeruli of kidney.

Disorder	Pathology / Key Cause	Clinical Features & Management
Uremia	Kidney failure leading to urea buildup in blood	Toxic urea accumulation; managed via Hemodialysis or transplant
Renal Calculi	Precipitation of insoluble salt crystals (calcium oxalates)	Severe pain (renal colic); obstructs urine flow in kidney/ureter
Glomerulonephritis	Immune-mediated inflammation of glomeruli	Hematuria, proteinuria, hypertension, reduced GFR
Diabetes Mellitus	Insulin deficiency / resistance leading to hyperglycemia	Glycosuria and Ketonuria detected in urine examination

FINAL REVISION — ULTRA-QUICK EXAM FLASHCARDS

10-Minute High-Yield Review

★ GFR & Ultrafiltration

GFR = 125 mL/min = 180 L/day. 99% filtrate reabsorbed. Podocytes leave filtration slits in Bowman's capsule.

★ PCT Epithelium

Simple cuboidal brush border epithelium. Reabsorbs 70–80% electrolytes, H₂O, and all glucose/amino acids.

★ Henle's Loop Permeability

Descending = Permeable to H₂O, impermeable to NaCl.

Ascending = Impermeable to H₂O, transports NaCl.

★ Hormonal Trio

ADH (water retention) + RAAS/Renin-Aldosterone (Na⁺/H₂O retention, raises BP) + ANF (vasodilation, lowers BP).

COMPLETE NCERT TEXTBOOK EXERCISE SOLUTIONS

All 12 Textbook Questions Solved with Accurate NCERT Academic Language

Answers Verified

Q1. Define Glomerular Filtration Rate (GFR).

Answer: Glomerular Filtration Rate (GFR) is defined as the total amount of filtrate formed by the glomeruli of both kidneys per minute. In a healthy adult human individual, the normal GFR is approximately **125 mL/minute**, which amounts to roughly **180 Litres per day**.

Q2. Explain the autoregulatory mechanism of GFR.

Answer: Kidneys possess a built-in autoregulatory feedback mechanism to regulate GFR, primarily carried out by the **Juxtaglomerular Apparatus (JGA)**:

- JGA is a specialized cellular modification formed at the point of contact between the Distal Convolved Tubule (DCT) and the afferent arteriole.
- A fall in GFR or glomerular blood pressure activates the Juxtaglomerular (JG) cells to release the enzyme **Renin**.
- Renin stimulates conversion of angiotensinogen to Angiotensin II, which constricts arterioles and increases glomerular blood flow, restoring GFR back to normal.

Q3. Indicate whether the following statements are true or false:

- (a) Micturition is carried out by a reflex. → **TRUE** (Micturition reflex via CNS motor signals).
- (b) ADH helps in water elimination, making the urine hypotonic. → **FALSE** (ADH promotes water *reabsorption*, making urine hypertonic/concentrated).
- (c) Protein-free fluid is filtered from blood plasma into the Bowman's capsule. → **TRUE** (Proteins are retained in plasma during ultrafiltration).
- (d) Henle's loop plays an important role in concentrating the urine. → **TRUE** (Creates hyperosmolar medullary gradient).
- (e) Glucose is actively reabsorbed in the proximal convoluted tubule. → **TRUE** (Reabsorbed actively against concentration gradient).

Q4. Give a brief account of the counter current mechanism.

Answer: The counter-current mechanism operates between the two limbs of the Loop of Henle and the two limbs of Vasa Recta to produce concentrated urine:

1. Flow of filtrate in Henle's loop limbs and flow of blood in Vasa recta limbs run in opposite directions (counter-current).
2. NaCl is transported out by ascending limb of Henle's loop and enters descending limb of Vasa Recta, and is returned to medullary interstitium by ascending Vasa Recta.
3. Small amounts of urea enter thin ascending limb and are returned to interstitium by collecting duct.
4. This maintains an increasing osmolarity gradient from **300 mOsmol L⁻¹ in cortex to 1200 mOsmol L⁻¹ in inner medulla**, enabling easy osmotic extraction of water from collecting duct to concentrate urine 4x.

Q5. Describe the role of liver, lungs and skin in excretion.

Answer:

- **Lungs:** Excrete approximately 200 mL of CO₂ per minute along with significant quantities of water vapor during respiration.
- **Liver:** Secretes bile containing bilirubin, biliverdin, cholesterol, degraded steroid hormones, vitamins, and drugs, which pass out with digestive wastes.
- **Skin:** Sweat glands eliminate sweat (containing NaCl, small amounts of urea, lactic acid). Sebaceous glands eliminate sterols, hydrocarbons, and waxes via sebum.

Q6. Explain micturition.

Answer: Micturition is the voluntary/involuntary process of release of urine from the urinary bladder through the urethra. As urine accumulates, stretch receptors on bladder walls send sensory signals to the CNS. The CNS responds with motor signals triggering simultaneous contraction of smooth detrusor muscles of the bladder and relaxation of the urethral sphincter, causing expulsion of urine via the micturition reflex.

Q7. Match the items of column I with those of column II:

Column I	Column II (Matched Solution)
(a) Ammonotelism	(iii) Bony fish
(b) Bowman's capsule	(v) Renal tubule
(c) Micturition	(iv) Urinary bladder
(d) Uricotelism	(i) Birds
(e) ADH	(ii) Water reabsorption

Q8. What is meant by the term osmoregulation?

Answer: Osmoregulation is the physiological process by which an organism maintains the optimal balance of water content, total solute concentration, and osmotic pressure in its body fluids to ensure homeostatic cellular function.

Q9. Terrestrial animals are generally either ureotelic or uricotelic, not ammonotelic. Why?

Answer: Ammonia is highly toxic and requires a very large volume of water for its elimination (it is excreted by diffusion across surfaces in aquatic environments). Terrestrial habitats face water scarcity. Therefore, terrestrial animals convert toxic ammonia into less toxic urea (ureotelism) or non-toxic insoluble uric acid (uricotelism) to conserve body water.

Q10. What is the significance of juxta glomerular apparatus (JGA) in kidney function?

Answer: JGA acts as a crucial autoregulatory sensor in the kidney. When blood pressure or GFR drops, JGA secretes Renin, initiating the Renin-Angiotensin-Aldosterone System (RAAS). Angiotensin II constricts blood vessels and Aldosterone increases Na⁺/water reabsorption, thereby restoring normal blood pressure and GFR.

Q11. Name the following:

- (a) A chordate animal having flame cells as excretory structures: → **Amphioxus (Cephalochordata)**
- (b) Cortical portions projecting between the medullary pyramids in human kidney: → **Columns of Bertini (Renal Columns)**
- (c) A loop of capillary running parallel to the Henle's loop: → **Vasa Recta**

Q12. Fill in the gaps:

- (a) Ascending limb of Henle's loop is **impermeable** to water whereas the descending limb is **permeable** to it.
- (b) Reabsorption of water from distal parts of the tubules is facilitated by hormone **Vasopressin / ADH**.
- (c) Dialysis fluid contains all the constituents as in plasma except **nitrogenous wastes (urea)**.
- (d) A healthy adult human excretes (on an average) **25 – 30** gm of urea/day.